

IT Workshop: Test 1

1. What is the output of the following?

```
x = ['ab', 'cd']
for i in x:
    i.upper()
print(x)
```

- a) ['ab', 'cd'].
- b) ['AB', 'CD'].
- c) [None, None].
- d) none of the mentioned

2. What is the output of the following?

```
i = 1
while True:
    if i%7 == 0:
        break
    print(i)
    i += 1
```

- a) 1 2 3 4 5 6
- b) 1 2 3 4 5 6 7
- c) error
- d) none of the mentioned

3. Which of the following commands will create a list?

- a) list1 = list()
- b) list1 = [].
- c) list1 = list([1, 2, 3])
- d) all of the mentioned

4. What is the output when we execute list("hello")?

- a) ['h', 'e', 'l', 'l', 'o'].
- b) ['hello'].
- c) ['llo'].
- d) ['olleh'].

5. What is the output of the following?

```
x = [i**+1 for i in range(3)];
print(x);
```

- a) [0, 1, 2].
- b) [1, 2, 5].
- c) error, **+ is not a valid operator
- d) error, ';' is not allowed

6. What is the output of the following?

```
k=[print(i) for i in my_string if i
not in "aeiou"]
```

- a) prints all the vowels in my_string
- b) prints all the consonants in my_string
- c) prints all characters of my_string that aren't vowels
- d) prints only on executing print(k).

7. Which of the following is a Python tuple?

- a) [1, 2, 3].
- b) (1, 2, 3)
- c) {1, 2, 3}
- d) {}

8. Suppose t = (1, 2, 4, 3), which of the following is incorrect?

- a) print(t[3])
- b) t[3] = 45
- c) print(max(t))
- d) print(len(t))

9. Which of these about a set is not true?

- a) Mutable data type
- b) Does not allows duplicate values
- c) Data type with unordered values
- d) Immutable data type

10. Which of the following is not the correct syntax for creating a set?

- a) set([[1,2],[3,4]])
- b) set([1,2,2,3,4])
- c) set((1,2,3,4))
- d) {1,2,3,4}

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11. Which of the following statements create a dictionary?

- a) `d = {}`
- b) `d = {"john":40, "peter":45}`
- c) `d = {40:"john", 45:"peter"}`
- d) All of the mentioned

12. Read the code shown below carefully and pick out the keys?

```
d = {"john":40, "peter":45}
```

- a) "john", 40, 45, and "peter"
- b) "john" and "peter"
- c) 40 and 45
- d) `d = (40:"john", 45:"peter")`

13. What Will Be The Output Of The Following Code Snippet?

```
a=[1,2,3,4,5,6,7,8,9]
a[::2]=10,20,30,40,50,60
print(a)
```

- a) ValueError: attempt to assign sequence of size 6 to extended slice of size 5
- b) [10, 2, 20, 4, 30, 6, 40, 8, 50, 60]
- c) [1, 2, 10, 20, 30, 40, 50, 60]
- d) [1, 10, 3, 20, 5, 30, 7, 40, 9, 50, 60]

14. What is the output of the following?

```
x = ['ab', 'cd']
for i in x:
    x.append(i.upper())
print(x)
```

- a) ['AB', 'CD'].
- b) ['ab', 'cd', 'AB', 'CD'].
- c) ['ab', 'cd'].
- d) none of the mentioned

15. What is the output of the following?

```
x = 2
for i in range(x):
    x -= 2
    print (x)
```

- a) 0 1 2 3 4 ...
- b) 0 -2
- c) 0
- d) error

16. What is the output of the following code?

```
places = ['Bangalore', 'Mumbai', 'Delhi']
places1 = places
places2 = places[:]
places1[1]="Pune"
places2[2]="Hyderabad"
print(places)
```

- a) ['Bangalore', 'Pune', 'Hyderabad'].
- b) ['Bangalore', 'Pune', 'Delhi'].
- c) ['Bangalore', 'Mumbai', 'Delhi'].
- d) ['Bangalore', 'Mumbai', 'Hyderabad'].

17. What is the output of the following code?

```
a=165
b=sum(list(map(int,str(a))))
print(b)
```

- a) 561
- b) 5
- c) 12
- d) Syntax error

18. What is the output of the following code?

```
a=[13,56,17]
a.append([87])
a.extend([45,67])
print(a)
```

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- a) [13, 56, 17, [87], 45, 67].
- b) [13, 56, 17, 87, 45, 67].
- c) [13, 56, 17, 87, [45, 67]].
- d) [13, 56, 17, [87], [45, 67]].

19. What is the output of the following code?

```
a=(1,2,(4,5))
b=(1,2,(3,4))
print(a<b)
```

- a) False
- b) True
- c) Error, < operator is not valid for tuples
- d) Error, < operator is valid for tuples but not if there are sub-tuples

20. What is the output of the following piece of code when executed in Python shell?

```
a=("Check")*3
print(a)
```

- a) ('Check','Check','Check')
- b) * Operator not valid for tuples
- c) 'CheckCheckCheck'
- d) Syntax error

21. What is the output of the code shown below?

```
z=set('abc')
z.add('san')
z.update(set(['p','q']))
print(z)
```

- a) {'abc','p','q','san'}
- b) {'a','b','c',['p','q'],'san'}
- c) {'a','c','c','p','q','s','a','n'}
- d) {'a','b','c','p','q','san'}

22. What is the output of the code shown below?

```
s=set([1,2,3])
print(s.union([4,5]))
print(s|([4,5]))
```

- a) {1, 2, 3, 4, 5}
- {1, 2, 3, 4, 5}
- b) Error
- {1, 2, 3, 4, 5}
- c) {1, 2, 3, 4, 5}
- Error
- d) Error
- Error

23. What is the output of the following?

```
print([i+j for i in "abc" for j in
"def"])
```

- a) ['da','ea','fa','db','eb','fb','dc','ec','fc'].
- b) [['ad','bd','cd'],['ae','be','ce'],['af','bf','cf']].
- c) [['da','db','dc'],['ea','eb','ec'],['fa','fb','fc']].
- d) ['ad','ae','af','bd','be','bf','cd','ce','cf'].

24. What is the output of the code shown below?

```
l1=[1,2,3]
l2=[4,5,6]
[x*y for x in l1 for y in l2]
```

- a) [4, 8, 12, 5, 10, 15, 6, 12, 18]
- b) [4, 10, 18]
- c) [4, 5, 6, 8, 10, 12, 12, 15, 18]
- d) [18, 12, 6, 15, 10, 5, 12, 8, 4]

25. What is the output of the following snippet of code?

```
a = {}
a[1] = 1
a['1'] = 2
a[1]=a[1]+1
count = 0
for i in a:
    count += a[i]
print(count)
```

- a) 1
- b) 2
- c) 4
- d) Error, the keys can't be a mixture of letters and numbers.

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26. What is the output of the following snippet of code? 30. What is the output of the following?

```
test = {1:'A', 2:'B', 3:'C'}
test = {}
print(len(test))
```

- a) 0
- b) None
- c) 3
- d) An exception is thrown

27. What Will Be The Output Of The Following Code Snippet?

```
arr = [[1, 2, 3, 4],
       [4, 5, 6, 7],
       [8, 9, 10, 11],
       [12, 13, 14, 15]]
for i in range(0, 4):
    print(arr[i].pop())
```

- a) 1 2 3 4
- b) 1 4 8 12
- c) 4 7 11 15
- d) 12,13,14,15

28. What Will Be The Output Of The Following Code Snippet?

```
arr = [1, 2, 3, 4, 5, 6]
for i in range(1, 6):
    arr[i - 1] = arr[i]
for i in range(0, 6):
    print(arr[i], end = " ")
```

- a) 1 2 3 4 5 6
- b) 2 3 4 5 6 1
- c) 1 1 2 3 4 5
- d) 2 3 4 5 6 6

29. What is the output of the following?

```
for i in 'abcd'[::-1]:
    print (i)
```

- a) a b c d b) d c b a
- c) error d) none of the mentioned

```
for i in '':
    print (i)
```

- a) None
- b) (nothing is printed)
- c) error
- d) none of the mentioned.

31. What is the output of the following code?

```
a=[[]]*3
a[1].append(7)
print(a)
```

- a) Syntax error
- b) [[7], [7], [7]].
- c) [[7], [], []].
- d) [[], 7, [], []].

32. What is the output of the following code?

```
lst=[3,4,6,1,2]
lst[1:2]=[7,8]
print(lst)
```

- a) [3, 7, 8, 6, 1, 2].
- b) Syntax error
- c) [3,[7,8],6,1,2].
- d) [3,4,6,7,8].

33. Is the following piece of code valid?

```
a,b,c=1,2,3
print( a,b,c )
```

- a) Yes, [1,2,3] is printed
- b) No, invalid syntax
- c) Yes, 1,2,3 is printed
- d) 1 is printed

34. What is the output of the following piece of code when executed in Python shell?

```
a=(1,2)
b=(3,4)
c=a+b
print(c)
```

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- a) (4,6)
- b) (1,2,3,4)
- c) Error as tuples are immutable
- d) None

35. What is the output of the following piece of code?

```
a = ('check',)
n = 2
for i in range(int(n)):
    a = (a,)
    print(a)
```

- a) Error, tuples are immutable
- b) (('check',),)
 - ((('check',),),).
- c) (('check',)'check',)
- d) (('check',)'check',)
 - ((('check',)'check',)'check',)

36. What is the output of the line of code shown below?

```
s1= {1, 2, 3}
s1.issubset(s1)
```

- a) True
- b) Error
- c) No output
- d) False

37. What is the output of this code?

```
s1={1, 2, 3, 8}
s2={3, 4, 5, 6}
print(s1|s2)
print(s1.union(s2))
```

- a) {3}
 - {1, 2, 3, 4, 5, 6, 8}
- b) {1, 2, 4, 5, 6, 8}
 - {1, 2, 4, 5, 6, 8}
- c) {3}
 - {3}
- d) {1, 2, 3, 4, 5, 6, 8}
 - {1, 2, 3, 4, 5, 6, 8}

38. What is the output of the following?

```
my_string = "hello world"
k = [(i.upper(), len(i)) for i in
my_string]
print(k)
```

- a) [('HELLO', 5), ('WORLD', 5)].
- b) [('H', 1), ('E', 1), ('L', 1), ('L', 1), ('O', 1), (' ', 1), ('W', 1), ('O', 1), ('R', 1), ('L', 1), ('D', 1)].
- c) [('HELLO WORLD', 11)].
- d) none of the mentioned.

39. Which of the following matrices will throw an error in Python?

a) A = [[1, 2, 3],
[4, 5, 6],
[7, 8, 9]]

b) B = [[3, 3, 3]
[4, 4, 4]
[5, 5, 5]]

c) C = [(1, 2, 4),
(5, 6, 7),
(8, 9, 10)]

d) D = [2, 3, 4,
3, 3, 3,
4, 5, 6]

40. What is the output of the following code?

```
count={}
count[(1,2,4)] = 5
count[(4,2,1)] = 7
count[(1,2)] = 6
count[(4,2,1)] = 2
tot = 0
for i in count:
    tot=tot+count[i]
print(len(count)+tot)
```

- a) 25
- b) 17
- c) 16
- d) Tuples can't be made keys of a dictionary

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41. What is the output of the following code?

```
a={}
a[2]=1
a[1]=[2,3,4]
print(a[1][1])
```

- a) [2,3,4].
- b) 3
- c) 2
- d) An exception is thrown.

42. What Will Be The Output Of The Following Code Snippet?

```
box = {}
jars = {}
crates = {}
box['biscuit'] = 1
box['cake'] = 3
jars['jam'] = 4
crates['box'] = box
crates['jars'] = jars
print (len(crates[box]))
```

- a) 1
- b) 3
- c) 4
- d) Type Error

43. What is the output of the following?

```
i = 0
while i < 5:
    print(i)
    i += 1
    if i == 3:
        break
else:
    print(0)
```

- a) 0 1 2 0
- b) 0 1 2
- c) error
- d) none of the mentioned.

44. What is the output of the following?

```
x = "abcdef"
while i in x:
    print(i, end=" ")
```

- a) a b c d e f
- b) abcdef
- c) i i i i i ...
- d) error

45. What is the output of the following code?

```
a=[1,2,3,4]
b=[sum(a[0:x+1]) for x in
range(0,len(a)) ]
print(b)
```

- a) 10
- b) [1,3,5,7].
- c) 4
- d) [1,3,6,10].

46. What is the output of the following piece of code?

```
a=(2,3,1,5)
a.sort()
print(a)
```

- a) (1,2,3,5)
- b) (2,3,1,5)
- c) None
- d) Error, tuple has no attribute sort.

47. What is the output of the code shown below?

```
s1={3, 4}
s2={1, 2}
s3=set()
i=0
j=0
for i in s1:
    for j in s2:
        s3.add((i,j))
        i+=1
        j+=1
print(s3)
```

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- a) {(3, 4), (1, 2)}
- b) Error
- c) {(4, 2), (3, 1), (4, 1), (5, 2)}
- d) {(3, 1), (4, 2)}

48. Write a list comprehension for number and its cube for l=[1, 2, 3, 4, 5, 6, 7, 8, 9].

- a) [x**3 for x in l]
- b) [x^3 for x in l]
- c) [x**3 in l]
- d) [x^3 in l]

49. What is the output of the code shown below?

```
l=[[1 ,2, 3], [4, 5, 6], [7, 8, 9]]
[row[i] for row in l] for i in
range(3) ]
```

- a) Error
- b) [[1, 4, 7], [2, 5, 8], [3, 6, 9]]
- c) 1 4 7
2 5 8
3 6 9
- d) (1 4 7)
(2 5 8)
(3 6 9)

50. What is the output of the following piece of code?

```
a={ }
a.fromkeys ([1,2,3] ,"check")
```

- a) Syntax error
- b) {1:"check",2:"check",3:"check"}
- c) "check"
- d) {1:None,2:None,3:None}.